

# **Common Structures Across Physical Theories: A PIHR Meta-Theory of Recursive Organization**

## **From Green Functions and Multiple Scattering to Interference, Holonomy, and Quantum Networks**

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### **Abstract**

Different areas of physics often employ very different microscopic concepts, variables, equations, and experimental interpretations while nevertheless exhibiting similar forms of mathematical organization. Green-function methods, quantum scattering, relativistic bound-state equations, multiple-wave scattering, optical interference, resonant tunneling, gauge-dependent transport, and quantum networks all contain examples in which propagation between states is modified by local interactions, closed paths retain phase information, and repeated processes generate large or infinite families of histories. This work proposes PIHR as a structural meta-theory for describing such common organization. PIHR separates four roles: propagation, interaction, holonomy or closed-path monodromy, and recursive closure. The framework does not replace the underlying physical theories and does not identify systems with different microscopic meanings. Instead, it introduces an abstraction layer in which domain-specific quantities are mapped to common structural roles and compared at that level. A first-return factorization theorem is established for linear recursive systems, showing when arbitrarily many repeated histories can be represented exactly by a finite internal resolvent. Dyson resummation, Lippmann–Schwinger scattering, Bethe–Salpeter resummation, Foldy–Lax multiple scattering, Fabry–Pérot interference, quantum double-barrier tunneling, Aharonov–Bohm transport, and quantum graphs are then interpreted as distinct object-level realizations of a common meta-level organization. A three-arm multi-cycle quantum network provides a nontrivial quantitative test: the number of explicit histories grows exponentially with recursive depth while the exact representation remains three-dimensional. Classical controllability and observability criteria further determine when this finite representation is minimal. The resulting distinction between microscopic history complexity and structural realization complexity suggests that common architectures can exist across physical domains without implying equivalence of the underlying physics. PIHR is therefore proposed not as a new fundamental interaction or microscopic law, but as a testable meta-theoretical language for recurrent mathematical organization across physical theories and formalisms.

**Keywords:** structural meta-theory; theoretical physics; Green functions; recursive systems; multiple scattering; quantum networks; holonomy; resummation; quantum graphs; minimal realization; PIHR

### **1. Introduction**

Physics is divided into theories and formalisms that are distinguished by their physical objects, scales, dynamical laws, and experimental meanings. Quantum field theory,

nonrelativistic scattering theory, wave optics, mesoscopic transport, and network quantum mechanics are not interchangeable descriptions of the same physical system. Nevertheless, their mathematical formulations repeatedly contain related organizational motifs.

In quantum electrodynamics, Dyson showed how repeated irreducible contributions can be reorganized into dressed quantities [1]. Schwinger's Green-function formulation provided an operator language for propagation and field interactions [2], while Feynman's space-time formulation represented amplitudes through contributions associated with alternative histories [3]. These developments established several of the recurrent structures that remain central to modern theoretical physics.

The same organizational pattern appears in scattering and many-body problems. The Lippmann–Schwinger construction generates repeated propagation between potential interactions [4]. Foldy formulated multiple scattering of waves by distributed scatterers [5], and Lax developed a systematic multiple-scattering theory [6] together with its effective-field extension for dense systems [7]. Watson obtained a corresponding multiply scattered structure in the many-body problem [8]. These theories differ in physical interpretation, but each contains a finite set of elementary processes capable of generating arbitrarily long sequences.

Relativistic two-particle physics provides another example. The Bethe–Salpeter equation organizes repeated propagation between irreducible two-particle interaction kernels [9]. Repetition also appears outside Green-function and field-theoretical language. Fabry and Péro's interferometer produces its characteristic response through repeated internal reflections [10]. In semiconductor physics, finite-superlattice tunneling was analyzed by Tsu and Esaki [11], and resonant tunneling through semiconductor double barriers was demonstrated by Chang, Esaki, and Tsu [12].

Closed trajectories introduce a further structural element. Aharonov and Bohm demonstrated that closed-path phase information associated with the electromagnetic potential can influence interference [13]. Byers and Yang connected multiply connected geometries with flux-dependent quantum structure [14], and Aharonov–Bohm oscillations were subsequently observed in normal-metal rings [15]. Multi-channel transport through small rings can be formulated through scattering matrices [16], while the Fisher–Lee relation connects Green functions with transmission amplitudes [17]. Landauer's transport formulation provides another foundational connection between transmission and conductance [18].

Quantum graphs make this structural recurrence especially explicit. Kottos and Smilansky showed how quantum dynamics on graphs can be represented in terms of propagation and vertex scattering [19] and developed periodic-orbit descriptions of their spectra [20]. Kostykin and Schrader established general quantum-wire scattering on finite graphs and composition of graph scattering matrices [21]. Barra and Gaspard studied transport and dynamics on open graphs [22], while Kottos and Smilansky obtained exact scattering expressions for open chaotic graphs [23].

Closed-path information itself has several meanings. Wu and Yang formulated gauge structure using nonintegrable phase factors associated with paths [24]. Berry showed that cyclic adiabatic evolution generates geometric phase [25], and Wilczek and Zee extended this concept to non-Abelian holonomy [26]. Thus a closed cycle may carry ordinary dynamical phase, Abelian gauge phase, geometric phase, or a noncommuting matrix-valued holonomy, depending on the underlying theory.

A second body of mathematics is relevant to the present argument. Kalman showed that linear dynamical systems possess canonical structural decompositions [27] and developed the relation between state-space and input-output descriptions [28]. Ho and Kalman subsequently provided an effective construction of finite minimal realizations from input-output data [29]. These results establish that the complexity of a sequence of observable outputs need not equal the dimensionality of the smallest internal system capable of generating that sequence. Recursive representations in physics similarly permit infinite processes to be encoded through finite internal structures; the recursion method of Haydock, Heine, and Kelly provides a classic condensed-matter example [30].

These observations motivate a question different from the usual question posed within any one of these theories:

**Can recurrent forms of mathematical organization across physically different theories be described by a common theory at a higher structural level?**

The proposal developed here is that they can, at least for a restricted but broad class of linear recursive problems.

We introduce four structural roles:

$P$  = propagation,

$I$  = interaction or local transformation,

$H$  = holonomy or closed-path monodromy,

and

$R$  = recursive closure.

The framework will be called **PIHR**.

The central claim is not that these four quantities constitute new microscopic ingredients of nature. Nor is the claim that Dyson theory, scattering theory, optics, mesoscopic transport, and quantum graphs are physically equivalent.

Rather, PIHR is proposed as a **structural meta-theory**.

An object-level physical theory specifies physical states, fields, particles, interactions, observables, and dynamical laws. The PIHR meta-level instead asks which *structural role* a particular object performs inside the organization of that theory.

Thus,

$$G_0, \quad e^{ikL}, \quad G_{ij}$$

may be physically different objects but can all occupy a propagation role.

Likewise,

$$\Sigma, \quad V, \quad S_v$$

can occupy interaction roles even though a quantum-field self-energy, a scattering potential, and a vertex scattering matrix are physically different objects.

The proposed hierarchy is therefore

$$\boxed{\text{domain-specific physics} \rightarrow (P, I, H) \rightarrow K_{PIH} \rightarrow R(K_{PIH}).}$$

This paper develops that idea mathematically and tests it against established theories.

## 2. What Is Meant by a Structural Meta-Theory?

### 2.1 Object-level theories and meta-level structure

Let

$$\mathcal{D}$$

denote a physical theory, formalism, or model.

At the object level,  $\mathcal{D}$  specifies theory-dependent quantities such as Hamiltonians, propagators, wave amplitudes, self-energies, scattering matrices, fields, or Green functions.

PIHR does not alter these quantities.

Instead, define a structural map

$$\Phi_{\text{PIHR}}: \mathcal{D} \rightarrow \mathfrak{A}_{\text{PIHR}},$$

where

$$\mathfrak{A}_{\text{PIHR}}$$

is the space of PIHR representations.

The map assigns domain-specific mathematical objects to one of several structural roles.

Thus the meta-theory operates on **representations of physical processes**, not directly on physical reality.

### 2.2 The four structural roles

#### Propagation

$P$  transports an amplitude or state between internal events.

Examples include

$$G_0(E), \\ e^{ikL},$$

or a propagation matrix

$$P_{ij}.$$

#### Interaction

$I$  changes the propagating object locally.

Examples include

$$\Sigma, \quad V, \quad \mathcal{K}, \quad S_v.$$

### Holonomy or closed-path monodromy

$H$  stores information that becomes meaningful when a process traverses a closed cycle.

Examples include

$$e^{ikL_C}$$

for dynamical monodromy and

$$\exp\left(\frac{iq}{\hbar} \oint_C \mathbf{A} \cdot d\mathbf{l}\right)$$

for Abelian gauge holonomy.

### Recursive closure

$R$  generates the complete repeated process from a primitive return kernel.

For a bounded kernel  $K$ ,

$$R(K) = \sum_{n=0}^{\infty} K^n$$

where the series converges.

Hence

$$R(K) = (\mathbb{1} - K)^{-1}$$

under the usual Neumann-series condition.

The essential asymmetry is therefore

$$\boxed{P, I, H = \text{primitive structural data,}}$$

whereas

$$\boxed{R = \text{operation on the structure generated by } P, I, H.}$$

Consequently, the architecture is not best regarded as

$$P \oplus I \oplus H \oplus R.$$

Its more precise form is

$$\boxed{\mathcal{A} = R(P, I, H).}$$

### 3. PIHR Meta-Representation of a Physical System

Consider a domain-specific system  $\mathcal{D}$ .

Let

$$\mathcal{X}$$

be an internal state or boundary space.

Let

$$E: \mathcal{X}_{\text{in}} \rightarrow \mathcal{X}$$

be an entrance map and

$$F: \mathcal{X} \rightarrow \mathcal{X}_{\text{out}}$$

an exit map.

Let

$$\mathfrak{B} = \{\beta_1, \dots, \beta_m\}$$

be a finite set of primitive **first-return blocks**.

A first-return block begins in the internal boundary space, evolves according to the object-level theory, and returns to the internal boundary without an intermediate return to the chosen decomposition surface.

The amplitude of block  $\beta_\alpha$  is

$$B_\alpha = A(\beta_\alpha).$$

Each  $B_\alpha$  is composed from theory-specific representatives of the structural sectors:

$$B_\alpha \in \text{Alg}(P, I, H).$$

Define the first-return kernel

$$K_{PIH} = \sum_{\alpha=1}^m B_\alpha.$$

The meta-theoretical representation of  $\mathcal{D}$  is then

$$\Phi_{\text{PIHR}}(\mathcal{D}) = (\mathcal{X}, E, F, P, I, H, K_{PIH}, R).$$

Two theories can therefore differ completely in their physical interpretation while occupying analogous locations in the PIHR representation space.

## 4. PIH-Factorizable Systems

### Definition 1 — PIH-factorizable history system

A recursive physical history system is **PIH-factorizable** if every admissible non-direct history possesses a unique decomposition

$$\gamma = \gamma_{\text{out}} \circ \beta_{\alpha_n} \circ \dots \circ \beta_{\alpha_1} \circ \gamma_{\text{in}},$$

with

$$\beta_{\alpha_j} \in \mathfrak{B}.$$

The integer  $n$  is the recursive depth.

Unique factorization is essential.

Without it, two different block sequences could represent the same physical history, producing overcounting.

This condition defines the class for which the simplest form of PIHR closure is exact.

## 5. First-Return PIHR Resummation Theorem

### Theorem 1 — First-Return PIHR Resummation

Let  $\mathcal{X}$  be a Banach space.

Assume:

1. admissible histories are PIH-factorizable;
2. the primitive return family is finite;
3. amplitudes multiply under concatenation;
4. the return operators are bounded;
5. the first-return kernel is

$$K_{PIH} = \sum_{\alpha} B_{\alpha};$$

6. and

$$\| K_{PIH} \| < 1.$$

Then the contribution of all recursively generated histories is

$$\boxed{\mathcal{O}_{\text{rec}} = F(1 - K_{PIH})^{-1}E.}$$

If a nonrecursive direct term

$$\mathcal{O}_0$$

exists, then

$$\boxed{\mathcal{O} = \mathcal{O}_0 + F(1 - K_{PIH})^{-1}E.}$$

### Proof

Histories containing exactly  $n$  primitive returns contribute

$$\mathcal{O}^{(n)} = \sum_{\alpha_1, \dots, \alpha_n} F B_{\alpha_n} \cdots B_{\alpha_1} E.$$

By distributivity,

$$\mathcal{O}^{(n)} = F \left( \sum_{\alpha} B_{\alpha} \right)^n E.$$

Therefore

$$\mathcal{O}^{(n)} = F K_{PIH}^n E.$$

Summing over all recursive depths,

$$\mathcal{O}_{\text{rec}} = F \sum_{n=0}^{\infty} K_{PIH}^n E.$$

Because

$$\| K_{PIH} \| < 1,$$

the Neumann series converges and

$$\sum_{n=0}^{\infty} K_{PIH}^n = (\mathbb{1} - K_{PIH})^{-1}.$$

Hence

$$\mathcal{O}_{\text{rec}} = F(\mathbb{1} - K_{PIH})^{-1}E.$$

■

The theorem itself is elementary operator theory. Its proposed meta-theoretical significance is that the **same theorem acts on different object-level theories after their domain-specific quantities have been mapped into common structural roles.**

## 6. Resolvent Continuation and Pole Structure

The equality between a history sum and a Neumann series is asserted only where the series converges.

For a finite-dimensional kernel depending analytically on a spectral parameter  $z$ ,

$$K_{PIH} = K_{PIH}(z),$$

the matrix function

$$\mathcal{R}(z) = [\mathbb{1} - K_{PIH}(z)]^{-1}$$

can nevertheless be considered beyond the original convergence domain wherever the inverse exists.

Its poles satisfy

$$\boxed{\det[\mathbb{1} - K_{PIH}(z)] = 0.}$$

It is important to distinguish:

convergent path sum

from

analytic or meromorphic continuation of its resolvent representation.

PIHR does not identify these concepts automatically.

## 7. Structural Equivalence at the Meta-Level

A meta-theory requires a notion of equivalence different from physical identity.

## Definition 2 — PIHR structural equivalence

Let two systems possess representations

$$(\mathcal{X}_1, K_1, E_1, F_1)$$

and

$$(\mathcal{X}_2, K_2, E_2, F_2).$$

They are **PIHR-equivalent relative to their boundaries** if an invertible map

$$S: \mathcal{X}_1 \rightarrow \mathcal{X}_2$$

exists such that

$$K_2 = SK_1S^{-1},$$

$$E_2 = SE_1,$$

and

$$F_2 = F_1S^{-1}.$$

## Proposition 1

PIHR-equivalent systems have identical recursive boundary amplitudes.

### Proof

Since

$$K_2 = SK_1S^{-1},$$

we have

$$(\mathbb{1} - K_2)^{-1} = S(\mathbb{1} - K_1)^{-1}S^{-1}.$$

Therefore

$$F_2(\mathbb{1} - K_2)^{-1}E_2 = F_1(\mathbb{1} - K_1)^{-1}E_1.$$

■

This is a structural equivalence statement.

It does **not** imply that the physical observables, microscopic degrees of freedom, interpretation, dimensional units, or physical mechanisms of the two theories are identical.

## 8. Object-Level Realizations of the Meta-Structure

### 8.1 Dyson resummation

The Dyson equation has the form [1]

$$G = G_0 + G_0\Sigma G.$$

Iterating,

$$G = G_0 + G_0\Sigma G_0 + G_0\Sigma G_0\Sigma G_0 + \cdots.$$

Assign

$$P = G_0,$$

$$I = \Sigma,$$

$$H = \mathbb{1}.$$

The primitive return kernel can be chosen as

$$K_D = \Sigma G_0.$$

Thus

$$G = G_0 R(\Sigma G_0)$$

and

$$\boxed{G = G_0(\mathbb{1} - \Sigma G_0)^{-1}.}$$

The familiar result is not new.

The PIHR claim is that Dyson dressing instantiates the generic pattern

$$\boxed{P \rightarrow IP \rightarrow R(IP).}$$

## 8.2 Lippmann–Schwinger scattering

The transition operator satisfies [4]

$$T = V + V G_0^{(+)} T.$$

Iteration yields

$$T = V + V G_0^{(+)} V + V G_0^{(+)} V G_0^{(+)} V + \dots.$$

Set

$$P = G_0^{(+)}, \quad I = V, \quad H = \mathbb{1}.$$

Then

$$\boxed{T = IR(PI)}$$

or

$$\boxed{T = V(\mathbb{1} - G_0^{(+)} V)^{-1}.}$$

Dyson and Lippmann–Schwinger theory are therefore different object-level theories with a shared meta-level recursive organization.

## 8.3 Bethe–Salpeter resummation

Let

$$P_2 = G_1 G_2$$

denote two-particle propagation and

$$\mathcal{K}$$

an irreducible two-particle kernel [9].

An inhomogeneous Bethe–Salpeter form is

$$\Gamma = \mathcal{K} + \mathcal{K}P_2\Gamma.$$

Therefore

$$\Gamma = \mathcal{K} + \mathcal{K}P_2\mathcal{K} + \mathcal{K}P_2\mathcal{K}P_2\mathcal{K} + \cdots,$$

and

$$\boxed{\Gamma = IR(PI),}$$

where

$$I = \mathcal{K}, \quad P = P_2.$$

The underlying interaction kernel is physically very different from the potential in Lippmann–Schwinger theory, yet its recursive structural role is analogous.

### 8.4 Foldy–Lax multiple scattering

For  $N$  localized scatterers, write schematically [5–8]

$$u_i = u_i^{(0)} + \sum_{j \neq i} G_{ij} t_j u_j.$$

Define the propagation matrix

$$P_{ij} = (1 - \delta_{ij})G_{ij}$$

and the local interaction matrix

$$I = \text{diag}(t_1, \dots, t_N).$$

Then

$$\mathbf{u} = \mathbf{u}^{(0)} + P I \mathbf{u}.$$

Hence

$$\boxed{\mathbf{u} = (\mathbb{1} - P I)^{-1} \mathbf{u}^{(0)}.}$$

An arbitrarily long multiple-scattering sequence is represented by a finite matrix when the number of scatterers is finite.

## 9. Interference Systems as PIHR Realizations

### 9.1 Fabry–Pérot interferometer

For two partially reflecting mirrors separated by  $L$ , the single-pass propagation amplitude is

$$P = e^{ikL}.$$

Let the internal reflection amplitudes be

$$r_1', \quad r_2.$$

The round-trip kernel is

$$K_{\text{FP}} = r_1' r_2 e^{2ikL}.$$

The transmission amplitude becomes

$$t_{\text{FP}} = \frac{t_1 t_2 e^{ikL}}{1 - r_1' r_2 e^{2ikL}}.$$

Here the closed-path factor

$$e^{2ikL}$$

is ordinary dynamical monodromy rather than gauge holonomy.

## 9.2 Quantum double-barrier tunneling

For two barriers separated by  $L$ , the same repeated-reflection structure gives [11,12]

$$t_{\text{DB}} = \frac{t_1 t_2 e^{ikL}}{1 - r_1' r_2 e^{2ikL}}.$$

Thus the scalar kernels satisfy

$$K_{\text{FP}} \cong K_{\text{DB}}$$

after the corresponding amplitudes are identified.

This does not equate classical optics with quantum tunneling.

It states that their recursive **architectures** are isomorphic at this level.

## 10. Genuine Holonomy: Aharonov–Bohm Networks

The Aharonov–Bohm system changes the meaning of  $H$ .

For a closed cycle  $C$ ,

$$H(C) = \exp\left(\frac{iq}{\hbar} \oint_C \mathbf{A} \cdot d\mathbf{l}\right).$$

This phase is tied to the loop rather than to one local scattering event [13,14,24].

Consider a two-arm ring.

Let forward and reverse propagation be represented by

$$D_f = P_f H_f$$

and

$$D_b = P_b H_b.$$

Let junction scattering matrices be

$$J_L, \quad J_R.$$

A primitive return kernel is

$$K_{\text{AB}} = J_L D_b J_R D_f.$$

For entrance vector  $a$  and exit vector  $b^\dagger$ ,

$$t_{\text{AB}} = b^\dagger D_f (\mathbb{1} - K_{\text{AB}})^{-1} a.$$

Now all four roles are essential:

$P$  = arm propagation,  
 $I$  = junction scattering,  
 $H$  = gauge holonomy,  
 $R$  = sum over repeated windings.

## 11. General Quantum Graphs

For a graph with  $B$  internal bonds, one may work on the directed-bond space of dimension  $2B$  [19–23].

A one-step internal evolution operator has schematic form

$$U(k, \Phi) = S P(k) H(\Phi),$$

where operator ordering depends on convention.

Here,

$$S$$

contains vertex scattering,

$$P(k)$$

contains bond propagation, and

$$H(\Phi)$$

contains loop-dependent phase information.

The repeated internal dynamics is

$$\mathbb{1} + U + U^2 + \dots.$$

Hence

$$\boxed{R(U) = (\mathbb{1} - U)^{-1}.$$

A closed graph may have spectral condition

$$\boxed{\det[\mathbb{1} - U(k, \Phi)] = 0,}$$

while an open graph couples this internal resolvent to entrance and exit channels.

Quantum graphs therefore provide a natural object-level laboratory for the PIHR meta-theory.

## 12. Cross-Domain Structural Comparison

| Object-level problem | Propagation P            | Interaction I      | Holonomy/monodromy H | Recursive process R |
|----------------------|--------------------------|--------------------|----------------------|---------------------|
| Dyson equation       | bare propagator          | self-energy        | trivial              | repeated insertions |
| Lippmann–Schwinger   | free Green function      | potential          | trivial              | repeated scattering |
| Bethe–Salpeter       | two-particle propagation | irreducible kernel | usually trivial      | repeated kernels    |

|                       |                                 |                        |                                |                                      |
|-----------------------|---------------------------------|------------------------|--------------------------------|--------------------------------------|
| Foldy–Lax             | inter-scatterer<br>Green matrix | local scatterers       | trivial                        | multiple<br>scattering               |
| Fabry–Pérot           | cavity<br>propagation           | mirror<br>reflection   | dynamical round-<br>trip phase | repeated<br>reflections              |
| Double barrier        | free propagation                | barrier<br>scattering  | cavity phase                   | repeated<br>tunneling<br>reflections |
| Aharonov–Bohm<br>ring | arm propagation                 | junction<br>scattering | gauge holonomy                 | repeated<br>windings                 |
| Quantum graph         | bond<br>propagation             | vertex<br>scattering   | cycle<br>phase/holonomy        | repeated graph<br>paths              |

The table illustrates the central meta-theoretical statement:

different physical meanings can instantiate the same structural roles.

### 13. A Nontrivial Test: Three-Arm $\Theta$ -Network

A useful meta-theory must do more than rename familiar terms.

We therefore consider a system in which the number of explicit histories grows rapidly while the structural representation remains finite.

Let two junctions

$$L, \quad R$$

be connected by three internal arms.

Define

$$P = \text{diag}(e^{ikL_1}, e^{ikL_2}, e^{ikL_3}).$$

Let the forward holonomy matrix be

$$H = \text{diag}(e^{i\alpha_1}, e^{i\alpha_2}, e^{i\alpha_3}).$$

Then

$$D_f = PH,$$

while reverse traversal is represented by

$$D_b = PH^\dagger$$

for the corresponding orientation convention.

Let

$$S_L, \quad S_R$$

be the internal junction-scattering matrices.

A complete internal return has kernel

$$K_\theta = S_L D_b S_R D_f.$$

Thus

$$\boxed{K_\theta \in M_3(\mathbb{C})}.$$

#### 14. Closed Cycles and Independent Holonomies

The three-arm  $\theta$ -network contains two independent cycles.

The loop using arm  $i$  forward and arm  $j$  backward accumulates phase

$$e^{i(\alpha_i - \alpha_j)}.$$

For example, choose

$$\phi_1 = \alpha_1 - \alpha_2,$$

$$\phi_2 = \alpha_1 - \alpha_3.$$

The third difference follows from

$$\alpha_2 - \alpha_3 = \phi_2 - \phi_1.$$

A common transformation

$$\alpha_j \rightarrow \alpha_j + \chi$$

does not change any loop difference.

Thus the meta-theoretical  $H$  sector separates closed-cycle information from local propagation labels.

#### 15. Exponential History Growth

Assume all internal channel transitions are nonzero, so every arm choice is admissible.

Without a complete return, the initial propagation can use any of three arms:

$$N_0 = 3.$$

One complete return requires:

- one initial forward arm;
- one backward arm;
- one new forward arm.

Hence

$$N_1 = 3^3 = 27.$$

For  $n$  complete returns there are

$$n + 1$$

forward arm choices and

$$n$$

backward arm choices.

Therefore

$$\boxed{N_n = 3^{2n+1} = 3 \cdot 9^n}.$$

The asymptotic history-growth factor is

$$g_{\text{hist}} = \lim_{n \rightarrow \infty} \frac{N_{n+1}}{N_n} = 9.$$

The cumulative number through depth  $n$  is

$$N_{\leq n} = \sum_{m=0}^n 3 \cdot 9^m$$

and therefore

$$N_{\leq n} = \frac{3(9^{n+1} - 1)}{8}.$$

At

$$n = 8,$$

this gives

$$N_{\leq 8} = 145\,282\,683.$$

The explicit history family is already enormous.

## 16. Exact Finite Representation

Despite this proliferation, all histories of depth  $n$  are represented coherently by

$$t^{(n)} = b^\dagger D_f K_\theta^n a.$$

Summing every recursive depth,

$$t = \sum_{n=0}^{\infty} b^\dagger D_f K_\theta^n a.$$

Therefore

$$t = b^\dagger D_f (1 - K_\theta)^{-1} a$$

where the path expansion converges.

The representation dimension is

$$d_{\text{PIHR}} = 3.$$

Thus

$$N_n \propto 9^n$$

while

$$d_{\text{PIHR}} = 3$$

for every  $n$ .

This produces a clean separation:

history-growth complexity  $\neq$  realization dimension.

## 17. Exponential-to-Finite Representation Proposition

### Proposition 2

For a completely mixing three-arm two-junction network, the number of distinct arm-sequence histories containing  $n$  complete internal returns is

$$N_n = 3 \cdot 9^n,$$

whereas their exact coherent amplitude is generated by a fixed three-dimensional return operator

$$K_\theta.$$

### Proof

The initial forward traversal has three alternatives.

Every complete return adds a backward arm and a subsequent forward arm, producing

$$3 \times 3 = 9$$

additional alternatives.

Hence

$$N_n = 3 \cdot 9^n.$$

Matrix multiplication of

$$K_\theta^n$$

sums over the same intermediate channel indices.

Therefore all histories at depth  $n$  are encoded in a fixed matrix power.

The dimension of the internal state space remains three.

■

The significance of Proposition 2 is not that matrix multiplication can sum paths; that is established mathematics.

Its relevance to PIHR is that the same meta-level representation distinguishes **combinatorial proliferation of histories** from **dimension of the structural generator**.

## 18. Minimal Structural Realization

A finite representation need not be minimal.

To test whether the three-dimensional PIHR representation contains redundant internal states, introduce a formal recursion variable  $z$  and define

$$\mathcal{T}(z) = F(1 - zK)^{-1}E.$$

Then

$$\mathcal{T}(z) = \sum_{n=0}^{\infty} z^n FK^nE.$$

The coefficients

$$FK^nE$$

are the recursion-depth amplitudes.

For a three-dimensional realization define the controllability matrix

$$\mathcal{C} = (E \quad KE \quad K^2E)$$

and observability matrix

$$\mathcal{O} = \begin{pmatrix} F \\ FK \\ FK^2 \end{pmatrix}.$$

Classical realization theory [27–29] gives the following.

**Proposition 3 — Conditional minimality**

If

$$\text{rank} \mathcal{C} = 3$$

and

$$\text{rank} \mathcal{O} = 3,$$

then the three-dimensional PIHR representation is controllable and observable and is therefore a minimal finite-dimensional linear realization of the same recursion-generating input-output function.

Consequently,

$$\boxed{d_{\min} = 3.}$$

This is important for the meta-theory.

The statement

$$d = 3$$

would otherwise merely describe one chosen representation.

Under the rank conditions, it becomes a statement about the minimum internal dimension required to reproduce the same recursive boundary response.

## 19. Meta-Theoretical Quantities

The preceding construction suggests several quantities that belong naturally to the structural level rather than to one microscopic theory.

### 19.1 Return-kernel equivalence class

The similarity class

$$[K] = \{SKS^{-1}\}$$

characterizes the internal linear recursion independently of basis choice.

## 19.2 Minimal realization dimension

Define

$$\boxed{d_{\min}(\mathcal{D})}$$

as the dimension of a minimal finite PIHR realization when one exists.

This measures structural representation complexity, not the number of physical degrees of freedom of the full underlying theory.

## 19.3 History-growth factor

When the number of admissible histories has asymptotic exponential growth, define

$$\boxed{g_{\text{hist}} = \limsup_{n \rightarrow \infty} N_n^{1/n}.$$

For the three-arm network,

$$g_{\text{hist}} = 9.$$

## 19.4 Pole set

For parameter-dependent kernels,

$$\mathcal{P} = \{z: \det[\mathbb{1} - K(z)] = 0\}$$

provides another structural characteristic.

## 19.5 Holonomy content

The  $H$  sector records whether recursion is:

$$H = \mathbb{1},$$

or contains ordinary closed-path monodromy,

$$H = e^{i\theta_{\text{dyn}}},$$

or genuine Abelian/non-Abelian holonomy.

These distinctions remain visible even when the recursive algebra is otherwise similar.

## 20. What PIHR Does—and Does Not—Unify

The word **meta-theory** must not be interpreted as claiming that all physics has been reduced to four elementary physical processes.

PIHR does not claim that

$$\Sigma = V = S_v.$$

It claims only that these objects may perform an analogous structural role.

Likewise,

$$G_0$$

and

$$e^{ikL}$$

are not physically identical.

They are representatives of a propagation role in their respective theories.

Therefore the relation is

$$\boxed{\text{physical identity} \neq \text{structural-role equivalence.}}$$

This distinction is essential.

PIHR seeks common **organization**, not common microscopic ontology.

## 21. Three Levels of Description

The framework can be organized into three levels.

### Level I — Domain physics

At this level one has:

QFT,  
scattering theory,  
wave optics,  
mesoscopic transport,  
quantum graphs.

Each contains its own physical laws and interpretation.

### Level II — Structural roles

Domain objects are mapped to

$$P, \quad I, \quad H.$$

The primitive first-return kernel is constructed:

$$K_{PIH}.$$

### Level III — Meta-level properties

One studies properties such as

$$[K],$$
$$d_{\min},$$
$$\mathcal{G}_{\text{hist}},$$

holonomy structure, and pole conditions.

The meta-theory therefore has the hierarchy

$$\boxed{\text{physical theory} \rightarrow \text{structural representation} \rightarrow \text{meta-invariants.}}$$

## 22. Why Recursion Is Different from $P$ , $I$ , and $H$

One conceptual consequence of this construction is that the original four labels are not fully symmetric.

Propagation, interaction, and holonomy characterize primitive histories.

Recursive closure acts **on combinations of those primitives**.

Thus

$$R$$

belongs to a different logical level.

The more accurate structural expression is

$$\boxed{\mathcal{A} = R(P, I, H)}.$$

This observation explains why recursive closure appears in so many different mathematical forms:

- Dyson resummation;
- multiple-scattering expansion;
- repeated cavity reflection;
- repeated network winding;
- recursive Green-function construction.

The details of  $P$ ,  $I$ , and  $H$  vary.

The closure logic can remain the same.

### 23. A Hierarchy of the $H$ Sector

The cross-domain comparison reveals that closed-path information itself occurs at different levels.

#### Type 0: trivial loop sector

$$H = \mathbb{1}.$$

Examples include simple Dyson and Lippmann–Schwinger resummations when no topological phase is involved.

#### Type I: dynamical monodromy

$$H_{\text{dyn}} = e^{ikL_C}.$$

Fabry–Pérot and double-barrier systems belong naturally to this class.

#### Type II: Abelian gauge holonomy

$$H_A = \exp\left(\frac{iq}{\hbar} \oint_C \mathbf{A} \cdot d\mathbf{l}\right).$$

Aharonov–Bohm networks provide the canonical example.

#### Type III: non-Abelian holonomy

$$H(C) \in U(N)$$

with potentially

$$H(C_1)H(C_2) \neq H(C_2)H(C_1).$$

Wilczek–Zee-type structures motivate this extension [26].

Thus the meta-theory preserves distinctions that would be lost if all phase factors were simply absorbed into propagation.

## 24. Finite Architectural Closure Conjecture

The examples motivate a broader conjecture.

### Conjecture 1 — Finite PIHR Closure

Consider a linear physical system possessing:

1. finitely many internal boundary states;
2. multiplicative amplitudes under composition;
3. local interaction or scattering maps;
4. closed-path information representable on the internal state space;
5. a finite-memory first-return rule;
6. unique factorization into primitive return blocks.

Then the complete recursive boundary response admits a finite PIHR representation

$$\mathcal{O} = \mathcal{O}_0 + F(\mathbb{1} - K_{PIH})^{-1}E$$

away from its resolvent singularities.

The conjectural content is **not**

$$(\mathbb{1} - K)^{-1} = \sum K^n.$$

That identity is standard.

The conjectural content is that an appropriate **finite structural factorization exists for a broad physical class**.

## 25. Limits of the Meta-Theory

The framework has clear limits.

First, PIHR currently addresses linear recursive systems most naturally.

Second, a finite first-return state space may not exist.

Third, systems with genuine long-term memory may require an expanding state space.

Fourth, nonlinear recursion is not covered by Theorem 1.

Fifth, continuum quantum-field problems can involve unbounded operators, ultraviolet divergences, renormalization, and asymptotic rather than convergent perturbation series.

Sixth, an operator product may not possess a unique  $P/I/H$  decomposition.

Seventh, different structural decompositions may be possible for the same object-level theory.

Eighth, a finite representation may fail to provide any practical simplification.

These are not secondary issues. They determine whether PIHR is a useful meta-theory or only a new notation.

## 26. Falsifiability Criteria

The proposal can be tested.

PIHR becomes scientifically useful if:

1. the definitions of  $P$ ,  $I$ ,  $H$ , and  $R$  remain stable across domains;
2. the same factorization rules apply without problem-specific modification;
3. structurally equivalent systems can be identified before solving them independently;
4. nontrivial finite or minimal representations can be constructed systematically;
5. meta-level quantities such as  $d_{\min}$  reveal information not obvious in the object-level formulation;
6. the framework produces analytic or computational simplifications;
7. or new deductions become possible from cross-domain structural equivalence.

The program fails in its stronger form if:

1. every example requires a new definition of the sectors;
2. PIHR merely renames standard matrix multiplication;
3. no structural invariants emerge;
4. no useful reduction occurs;
5. or physically unrelated systems can be called equivalent only through arbitrary choices of representation.

These criteria make the proposal empirically and mathematically vulnerable rather than purely interpretive.

## 27. Non-Abelian Extension

Suppose the closed-path sector is matrix-valued:

$$H(C) \in U(N).$$

Then the order of cycle traversal matters.

For two cycles,

$$H(C_1)H(C_2) \neq H(C_2)H(C_1)$$

in general.

Primitive return blocks must therefore retain path ordering.

A kernel might take the form

$$K_{PIH} = \sum_{\alpha} P_{\alpha_1} I_{\alpha_1} H(C_{\alpha_1}) \cdots P_{\alpha_m} I_{\alpha_m} H(C_{\alpha_m}).$$

Recursive closure can remain

$$(\mathbb{1} - K_{PIH})^{-1}$$

for a linear bounded kernel, but the internal structural information is now noncommutative. This provides a considerably more stringent future test than scalar interference examples.

## 28. Beyond Linear Closure

The elementary PIHR theorem concerns linear recursion.

A nonlinear system may instead satisfy

$$X = X_0 + \mathcal{F}(X).$$

There is then no general reduction to

$$(\mathbb{1} - K)^{-1}.$$

A broader meta-theory would require a nonlinear closure operation

$$R_{NL}[\mathcal{F}].$$

Possible future directions include:

- fixed-point theory;
- nonlinear operator equations;
- functional Dyson–Schwinger hierarchies;
- renormalization-group recursion;
- and memory-dependent dynamical systems.

The present manuscript does not claim such an extension.

## 29. Discussion

The central proposal of this work is a change of level.

Traditional theoretical physics asks how a specific physical system evolves.

PIHR instead asks:

**What structural roles recur in the mathematical organization of different physical theories?**

This is why the term **meta-theory** is appropriate in the restricted sense used here.

The framework does not add another Hamiltonian, field, particle, interaction, or fundamental constant.

Instead, it maps domain-level mathematical objects into a common structural language.

At the object level,

$$G_0, \quad V, \quad \Sigma, \quad S_v, \quad e^{ikL}, \quad H(C)$$

have distinct meanings.

At the structural level, their organizational functions can be classified.

This leads to the central hierarchy

$$\boxed{\text{object-level theory} \rightarrow (P, I, H) \rightarrow K_{PIH} \rightarrow R(K_{PIH}).}$$

The comparison across established physical domains then reveals two different types of commonality.

The first is **recursive commonality**.

Dyson resummation, potential scattering, multiple scattering, cavity reflection, resonant tunneling, and graph winding all generate repeated histories from finite local rules.

The second is **closed-path commonality**.

Optical cavities, quantum rings, gauge systems, and network models all retain information associated with cycles, although the physical meaning of that information ranges from ordinary dynamical phase to gauge holonomy.

PIHR separates these common features without erasing their physical distinctions.

The  $\theta$ -network provides the most concrete test.

The explicit history count grows as

$$N_n = 3 \cdot 9^n.$$

Yet the recursive dynamics is generated in a three-dimensional internal state space.

The significant comparison is therefore not “many paths versus nine matrix entries.”

Rather, it is

$$\boxed{g_{\text{hist}} = 9}$$

versus

$$\boxed{d_{\text{PIHR}} = 3.}$$

These quantify different notions of complexity.

Moreover, if the controllability and observability matrices have full rank, realization theory implies

$$d_{\min} = 3.$$

The finite architecture then cannot be reduced further within the relevant class of linear input-output realizations.

This creates a possible bridge between the proposed meta-theory and established mathematical system theory.

A mature PIHR program would therefore not merely classify known equations.

It would attempt to compute structural quantities such as

$$d_{\min},$$

identify cross-domain equivalence classes,

$$[K],$$

and determine when an infinite-history problem belongs to a finite-recursive universality class.

That is a substantially more stringent objective than demonstrating that several known equations contain geometric series.

### 30. Conclusions

We have proposed PIHR as a structural meta-theory for recurrent organization across different areas of physics.

Its four labels denote:

$P$  = propagation,

$I$  = interaction,

$H$  = holonomy or closed-path monodromy,

$R$  = recursive closure.

The central conceptual result is that these four roles are not symmetric.

The primitive physical structure lies in

$$(P, I, H),$$

while recursion acts upon their resulting first-return kernel.

Thus

$$\boxed{\mathcal{A} = R(P, I, H).}$$

A first-return resummation theorem was established for PIH-factorizable linear systems:

$$\boxed{\mathcal{O} = \mathcal{O}_0 + F(1 - K_{PIH})^{-1}E.}$$

The same meta-level architecture was then identified in:

- Dyson Green-function resummation;
- Lippmann–Schwinger scattering;
- Bethe–Salpeter resummation;
- Foldy–Lax multiple scattering;
- Fabry–Pérot interference;
- quantum double-barrier tunneling;
- Aharonov–Bohm transport;
- and quantum graphs.

These systems remain physically different.

Their commonality lies in structural organization.

A three-arm multi-cycle quantum network provided a nontrivial quantitative example.

Its number of explicit histories grows as

$$N_n = 3 \cdot 9^n,$$

while its exact internal realization remains

$$d_{\text{PIHR}} = 3.$$

Under full controllability and observability,

$$d_{\text{min}} = 3.$$

The framework therefore distinguishes

history complexity

from

structural realization complexity.

This distinction may be one of the most useful consequences of the meta-theoretical perspective.

PIHR does not presently constitute a fundamental unification of physics.

It makes the more limited and testable proposal that **different physical theories can share recurring mathematical architectures and that these architectures can themselves become objects of theoretical study.**

If this program can be extended to more complicated networks, non-Abelian holonomies, interacting many-body systems, and finite closures of coupled Green-function hierarchies—and if genuinely new results follow from the cross-domain structural representation—then PIHR could develop from a classification scheme into a substantive theory of common organization across physical theories.

## Declarations

### Funding

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### Conflict of Interest

The author declares no conflict of interest.

### Data Availability

No datasets were generated or analyzed in this theoretical study.

### Code Availability

No numerical code was required for the analytical results presented here.

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